

Thoracic and chest pain

with KEVIN SINGER and JEFF BOYLE

*I want to apologise to my fans. I broke my rib and again I am out of the fight.
I really wanted to come back, I'm not sure if I will ever come back.*

Khabib Nurmagomedov, UFC, October 2015

THORACIC PAIN

As with neck pain (Chapter 23) and low back pain (Chapter 29), it is frequently difficult for the clinician to make a precise diagnosis in patients with pain in the region of the thoracic spine given the interplay between the thorax, upper limb, neck, low back, and the cardiorespiratory and visceral systems. The most common musculoskeletal problems are disorders of the thoracic intervertebral joints and the numerous rib articulations as this region of the spine contributes stability to the axial skeleton. The thorax never sleeps as, even at rest, this complex interplay of joints, ligaments and muscles moves with respiration.

Injury to the intervertebral disc, the facet joints (also named zygapophyseal joints) or other nociceptive structures (Chapters 5 and 6) of the thoracic spine may lead to local or referred pain. Clinical presentation of these often articular problems is varied. Pain and altered motion are the dominant

features when one or more intervertebral segments or rib joints are involved. There may be associated abnormalities of the paraspinal and periscapular muscles as well as increased neural mechanosensitivity (Chapter 6). Thoracic intervertebral joint problems frequently refer pain to the lateral or anterior chest wall. Prolapse of a thoracic intervertebral disc is rare in athletes; however, it may be under-reported given the often diffuse symptoms that arise.^{1, 2} Appropriate imaging studies are often necessary to rule out this diagnosis. The astute clinician keeps all diagnostic options open when considering the thoracic region.

In adolescents, the most common cause of thoracic region pain is Scheuermann's disease (Chapter 44), a disorder of the growth plates of the thoracic vertebral endplates associated with an accentuated lower thoracic kyphosis. A list of the causes of pain in the region of the thoracic spine is shown in Table 28.1. The surface,

Table 28.1 Causes of thoracic pain

Common	Less common	Not to be missed
Intervertebral joint sprain	Rib stress reaction/fracture	Cardiac causes
• Disc	Fracture of the rib posteriorly	Peptic ulcer
• Zygapophyseal joints	Thoracic disc prolapse	Tumour (e.g. carcinoma of the breast with bony metastasis)
Paraspinal muscle strain	T4 syndrome	Herpes zoster
Costotransverse joint sprain		
Costovertebral joint sprain		
Scheuermann's disease (adolescents)		
Postural imbalance of neck and shoulder and upper thoracic spine		

muscle and cross-sectional anatomy of this area are shown in Figure 28.1.

Vague symptoms must alert the astute clinician to consider the possibility of visceral origins as convergence of pain pathways may mimic somatic disorders (Chapters 5 and 6). The literature reporting occult pain presentations of thoracic-like pain encourages caution when assessing an individual with unresolved, at times over-investigated, symptoms.^{3, 4} Consider 'red flags' (see Chapter 7).

Assessment

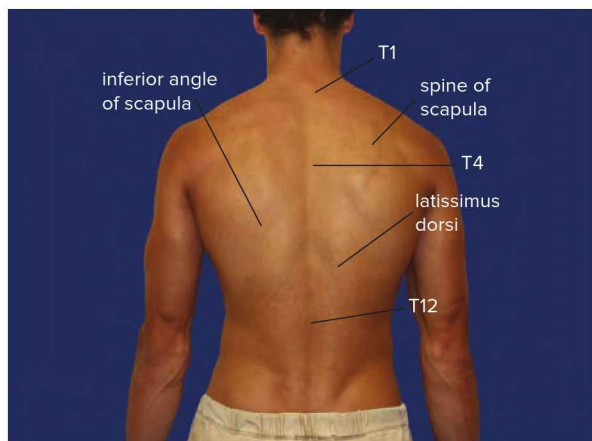
History

The patient often complains of pain between or around the scapulae. The pain may be central, unilateral or bilateral, and have commenced suddenly as a result of a specific movement or have been of gradual onset without discernible precipitating incident. Symptoms may be elicited by any active movement—particularly rotation (Fig. 28.2) or lateral flexion.⁵ The T4–8 segments show greater ranges of axial motion compared with the stiffer upper and lower segments. Note any associated sensory symptoms such as paraesthesia or numbness. Although dermatomal patterns are more predictable in the thoracic region, symptoms may depart from such conventions. Vague pain noted in the region of the shoulder may relate to disturbance of the cervicothoracic junction⁶ and, similarly, buttock, hip or inguinal region symptoms may have a low thoracic origin.⁵ The behaviour of symptoms relative to activity or diurnal variation will assist in classification of the condition (Fig. 28.3).

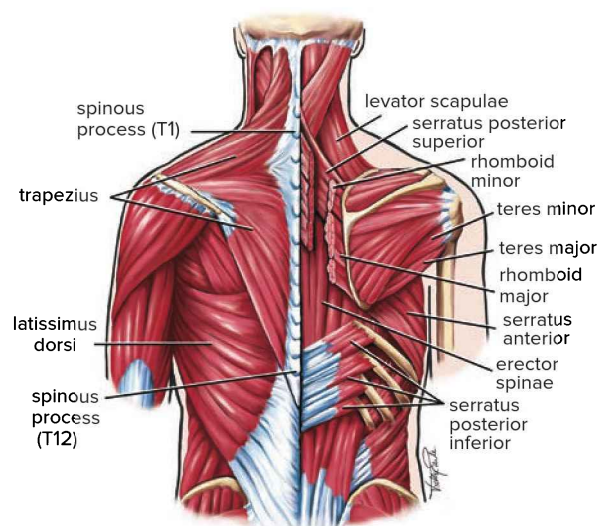
If you have established a musculoskeletal cause of symptoms, consider aspects of the clinical presentation relative to SINS (Severity, Irritability, Nature and Stage) to inform clinical reasoning and management. The Severity of the condition helps to determine the vigour of any prescribed activity or physical treatment; Irritability, or ease of aggravation, relates to the volume of prescribed activity or physical treatment; Nature refers to the type of condition (e.g. fracture, instability, neurological (radicular?, somatic?; see Chapter 6); and Stage means being relative to an acute, subacute or chronic classification.

Figure 28.1 Anatomy of the thoracic spine region

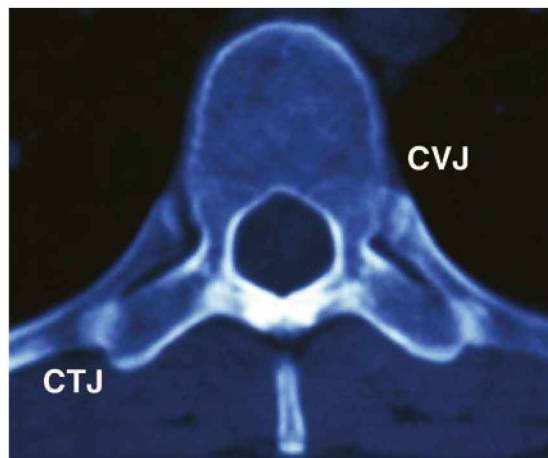
(a) Surface anatomy (b) Muscles of the thoracic spine region (c) Axial computed tomography (CT) image of the typical motion segment from the lower thoracic region. The most accessible rib articulation for palpation and mobilisation is the costotransverse joint (CTJ), with the costovertebral joint (CVJ) attached firmly to the lateral margin of the vertebral body, intervertebral disc and base of the pedicles



(a)



(b)



(c)

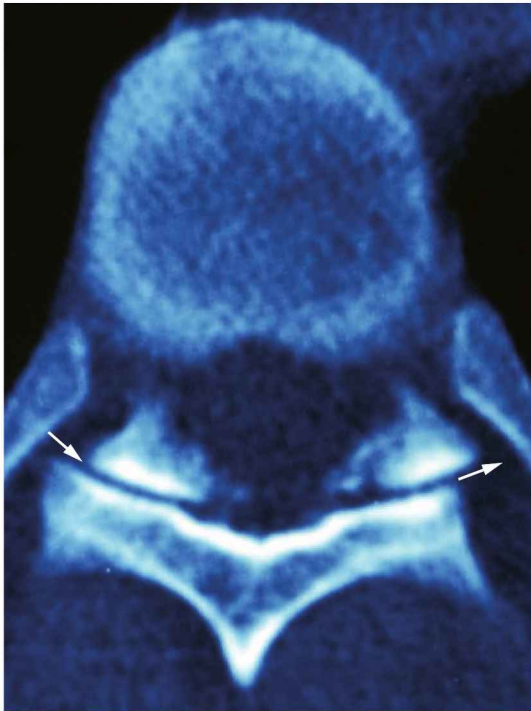


Figure 28.2 Axial CT depicting the nature of thoracic zygapophyseal joint translation in response to induced segmental rotation

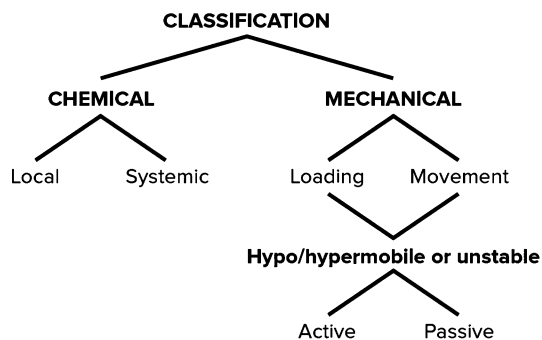


Figure 28.3 Flowchart of thoracic region disorders. Classification facilitates the differential diagnosis and better outcomes through prescription of condition-specific treatment.

In the thoracic region, carefully consider the potential for non-musculoskeletal causes. Red flags may include systemic illness, malignancy, unexplained weight loss, fever or night pain. As Professor Gwen Jull was fond of saying, 'If the voice in the back of your head tells you something is not right, listen to that voice' (and refer/investigate appropriately).



Figure 28.4 Examination of the patient with thoracic pain begins with observation and testing active movements

Physical examination

In the thoracic spine region, assess range of motion and mobility of each intervertebral segment and carefully palpate the paraspinal and periscapular soft tissue. Examine the lower cervical and upper lumbar spine and perform a neurological examination for altered mechanosensitivity.

1. Observation (Fig. 28.4)
 - a. any asymmetry, scoliosis or kyphosis should be noted
 - b. muscle spasm
 - c. skin changes (sweating or erythema may suggest an autonomic response)
2. Active movements
 - a. flexion
 - b. extension
 - c. rotation.

Active movements are assessed in the cardinal planes for range of motion, symptom reproduction, and aberrant patterns of movement that may be antalgic or indicate dysfunction. Combinations of these movements in two planes may help guide treatment (Fig. 28.5).⁷ A consistent pattern in which a movement that stretches the symptomatic area produces symptoms which are increased by the addition of movement in another plane that also has a stretch effect may suggest an extra-articular source of symptoms (myofascial, nerve, joint capsule), while a pattern consistent with symptoms primarily on compressive movements may indicate an intra-articular source. Mixed patterns may suggest multiple tissue involvement.

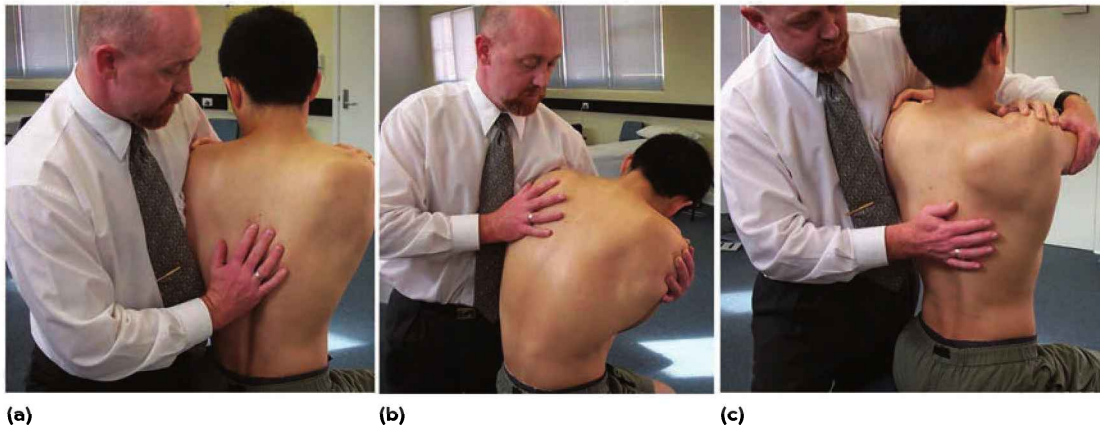


Figure 28.5 Combined movement assessment is performed by determining the most symptomatic primary movement; this movement is then repeated in the two extremes of another plane. For example, the primary symptomatic movement of axial plane left rotation (a) is subsequently performed in two positions in the sagittal plane—flexion (b) and extension (c)

3. Palpation

- a. soft tissue
- b. assess segmental unilateral mobility by posterior to anterior directional pressure over the transverse process (Fig. 28.6a)
- c. assess rib mobility by posterior to anterior directional pressure over the angle of the rib with the scapular protracted (Fig. 28.6b)
- d. zygapophyseal
- e. costovertebral joints and, indirectly, the costovertebral joints
- f. paraspinal muscles
- g. anterior palpation of sternal, chondral and clavicular joints.

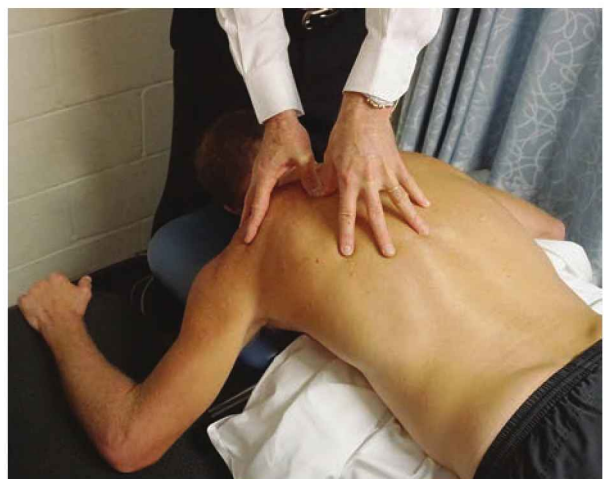
Soft tissue is palpated for tenderness, swelling, spasm, and trigger points while bony and chondral structures are assessed for tenderness and deformity and for motion anomalies.

4. Special tests

- a. springing of the ribs and costochondral junctions
- b. maximal inspiration
- c. cough/sneeze
- d. neural tissue sensitivity tests (slump test—Chapter 15)
- e. cervical spine examination (Chapter 23)
- f. lumbar spine examination (Chapter 29)
- g. muscle strength testing.



(a)



(b)

Figure 28.6 (a) Assess segmental unilateral mobility by posterior to anterior directional pressure over the transverse process (b) Assess rib mobility by posterior to anterior directional pressure over the angle of the rib with the scapular protracted

Investigations

Plain radiograph of the thoracic spine region is not routinely indicated as it usually adds little to the clinical picture. It may, however, demonstrate the presence of intervertebral growth plate abnormalities, indicate Scheuermann's disease, or show the presence of a secondary neoplastic disorder affecting the thoracic spine or ribs. Computed tomography (CT) and magnetic resonance imaging (MRI) scans may be indicated to exclude serious pathology in the presence of red flags such as unexplained weight loss or nocturnal pain. These imaging modalities may confirm a thoracic intervertebral disc prolapse or help stage neoplastic disease. Atypical pain patterns unresponsive to routine management should also signal the need for investigation, even in the young athlete (see also Chapters 5 and 6).

Thoracic intervertebral joint disorders

Intervertebral joint injuries involving the intervertebral discs, rib articulations and zygapophyseal joints are the most common cause of pain in this region. They may be of sudden or gradual onset. Examination may reveal hypomobility of one or more intervertebral segments associated with local tenderness over the spinous processes, the zygapophyseal joints, the costotransverse joints or the surrounding paravertebral muscles. However on occasion, pain rather than stiffness may be the main presenting feature.

Treatment aims to restore full mobility by mobilisation or manipulation techniques. Passive overpressure (Fig 28.7) can be used to assess treatment response (feel for resistance and assess pain response). Soft tissue therapy and graded specific exercise may be required to restore normal function in the paravertebral and periscapular muscles. Passive techniques include digital ischaemic pressure and sustained myofascial tension. Trigger points in the paraspinal muscles are common and may be treated with appropriate soft tissue techniques. Use of dry needling techniques for trigger points may be used with caution; avoid complications such as pneumothorax.

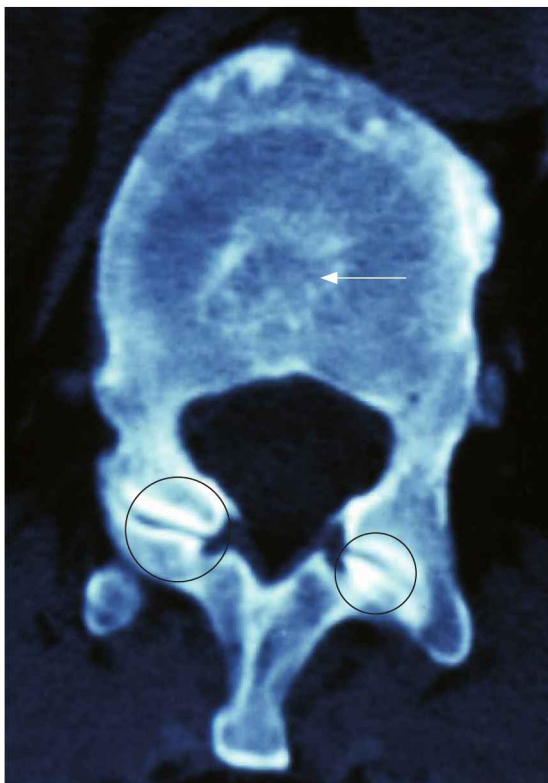
Note that there is considerable variation in joint anatomy at the transitional junctions of the spine, particularly at the thoracolumbar junction.⁸ These variations may contribute to subtle differences in segmental mechanics and injury patterns. While some zygapophyseal joints are remarkably asymmetrical, others may show a morphology that acts to constrain motion which should not be perceived as unusual hypomobility (Fig. 28.8). However, the transitional segments between the thoracic and lumbar regions are disposed to severe strain or injury during overload and these segments account for the highest incidence of spine-related trauma, particularly from high-energy sports (e.g. skiing, equestrian events, jet ski, motorcross).⁹



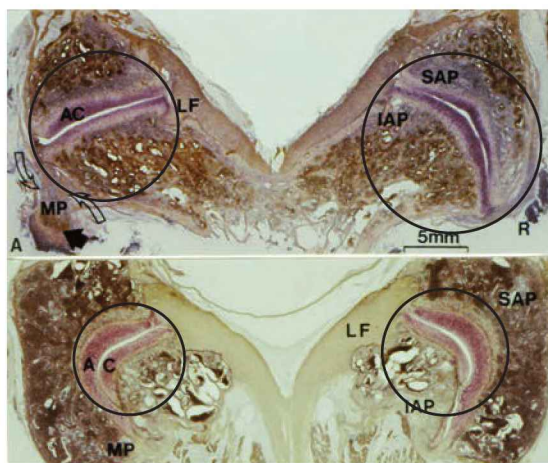
Figure 28.7 Thoracic rotation overpressure applied passively at the end of active range to assess for resistance and pain response

Costovertebral and costotransverse joint disorders

Disorders of the rib articulations include inflammatory spondyloarthropathies (such as ankylosing spondylitis), degenerative change (such as osteoarthritis) and mechanical joint sprains. Costotransverse joint problems are associated with localised tenderness and restricted mobility of the joints. This is often evident on deep inspiration and active movement as the ribs rotate in a predictable pattern relative to thoracic motion. Treatment may include mobilisation of the costotransverse joints, which will also have a modest influence on the deeper costovertebral articulations. There is no quality evidence to support image-guided corticosteroid injection as treatment when the condition remains unresponsive to standard management. A common problem of these small synovial joints may be derangement of intra-articular synovial fold inclusions which occupy the periphery of the internal joint cavity (Fig. 28.9). These conditions are usually amenable to mobilisations or manipulation.¹⁰



(a)



(b)

Figure 28.8 Zygapophyseal joint abnormalities (a) Axial CT scan highlighting marked zygapophyseal joint tropism (rotation in response to load) (circled); a central Schmorl's node is noted within the endplate (arrow) (b) Horizontal histological section (100 μ m) through the zygapophyseal joints of the thoracolumbar junction to demonstrate variations in articular morphology from asymmetric (top image) to an enclosing morphology which constrains motion (lower image)

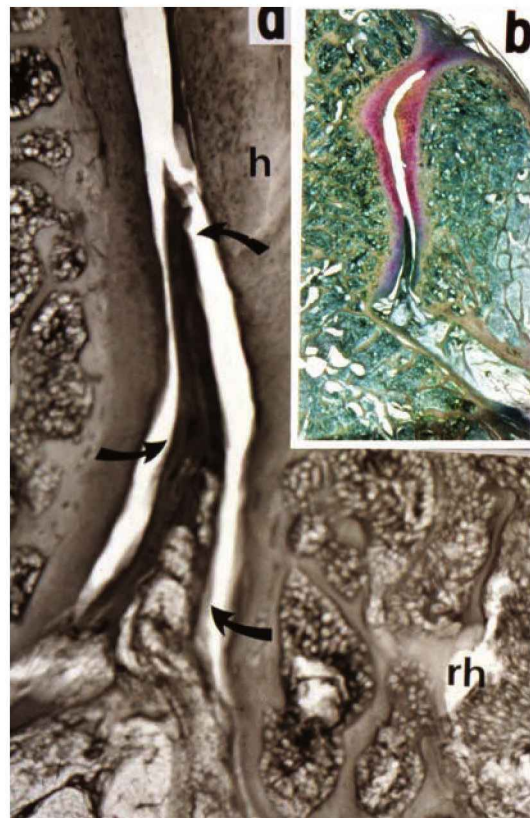


Figure 28.9 Painful costovertebral joint synovial inclusions (black arrows). Horizontal histological section through the dorsal region of a costovertebral joint demonstrates a long fibro-fatty synovial inclusion within the joint cavity. Entrapped innervated inclusions may contribute to localised thoracic pain which can often be relieved through manipulation

Scheuermann's disease

Scheuermann's disease is the most common cause of pain in the thoracic spine region in adolescents and is characterised by an accentuated low thoracic kyphosis arising from multiple vertebral endplate irregularities involving four or more vertebral bodies (Fig. 28.10). This condition is described in Chapter 44. Accentuated kyphosis may also arise from habitual training postures which involve loading into flexion.^{11, 3} Extended training periods in one posture (e.g. cycling) tend to be associated with adaptive changes, and modification to training postures may need to be considered when recommending long-term management.

Thoracic intervertebral disc prolapse

Prolapse of a thoracic intervertebral disc is a relatively rare condition that may be under-reported in the athletic



Figure 28.10 Multiple endplate lesions (Schmorl's nodes) evident within the lower thoracic vertebral bodies (circles) typical of Scheuermann's disease



(a)



(b)

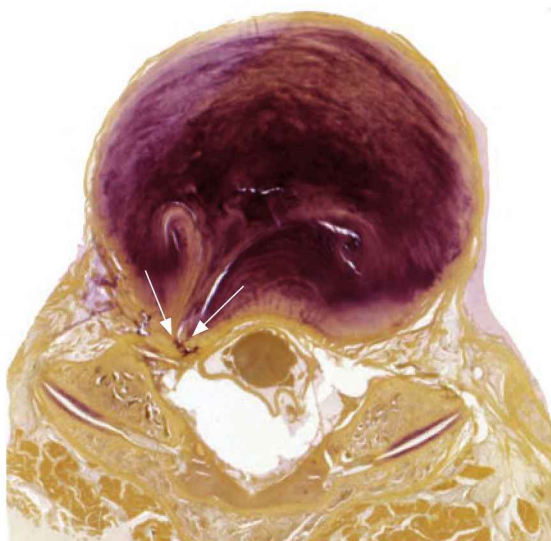


(c)

Figure 28.11 MRI of T12–L1 disc prolapse in a 26-year-old Australian rules footballer (a) T1-weighted axial MRI demonstrating left para-central displaced disc fragment (arrow) (b) T2-weighted axial MRI highlighting internal disc derangement (arrowheads) and disc herniation (arrow) (c) T2-weighted sagittal image showing disc fragment (circled) at T12–L1. It impinges on the spinal cord

community.^{1, 12} The segments that tend to be most commonly involved are the larger discs of the lower thoracic segments. MR images are shown for a T12–L1 disc prolapse (Fig. 28.11). The pathology of thoracic disc prolapse is shown in Figure 28.12. The clinical

B



(a)



(b)

Figure 28.12 Thoracic disc prolapse (a) Haematoxylin and eosin (H&E) stained horizontal section at T10–11 depicting a posterolateral prolapse of the intervertebral disc (arrows). Such prolapses are most common in the lower thoracic segments given the greater volume and height of these discs and loading patterns involving flexion and rotation (b) Macroscopic horizontal section of a T11–12 disc to demonstrate a midline annular fissure and small central prolapse (arrow), which is a common presentation for thoracic disc lesions

presentation involves local back pain with radicular pain radiating in the distribution of the affected thoracic spinal nerve/s. However, pain from a thoracic disc prolapse often does not follow a characteristic radicular pattern so this may confuse the diagnosis (see also Chapters 5 and 6).^{2, 13} Consider this potential diagnosis and refer for axial imaging and relevant pain intervention if you suspect the diagnosis.

T4 syndrome

Occasionally, patients present with diffuse arm pain and sensory symptoms such as bilateral paraesthesias in the hands, generalised headache and interscapular tightness which is attributed to intervertebral joint problems around the upper thoracic region. This vague constellation of symptoms has been labelled the T4 syndrome.¹⁴ Although not verified clinically, it speculates that the sympathetic nervous system contributes to symptoms. Examination often reveals hypomobility of the upper to middle thoracic segments together with a forward head position and shoulder protraction. Restoration of sagittal postural alignment of the head in relation to the thorax, shoulder retraction, improved neck and scapular muscle endurance, and mobilisation or manipulation of affected joints may relieve symptoms.

Postural imbalance of the neck, shoulder and upper thoracic spine

Common muscle imbalances relative to shoulder dysfunction include weakness of the scapula retractors, particularly inferior and middle trapezius, and serratus anterior. Stronger muscles will often exhibit increased tone and decreased length, and commonly include scalenus posterior, pectoralis minor and both rhomboid major and minor.¹⁵ These muscle imbalances may adversely influence segmental thoracic and rib articulation accessory motion.¹⁶

CHEST PAIN

Chest pain occurs not infrequently in athletes, usually due to musculoskeletal causes. In mature athletes, the possibility that pain is of cardiac origin must be considered. This possibility is increased in the presence of associated symptoms, such as palpitations or shortness of breath, or when there is a family history of cardiac disease. Male endurance athletes who perform high-intensity training are susceptible to atrial fibrillation, either as paroxysmal episodes or as a persistent form.¹⁷ Other causes of chest pain include peptic ulceration, gastroesophageal reflux, chest infection and malignancy.

The most common cause of chest pain in the athlete under 35 years is referred pain from the cervical or

Table 28.2 Causes of chest pain in the athlete

Common	Less common	Not to be missed
Rib trauma • Fracture • Contusion	Costochondritis	Cardiac causes
	Sternocostal joint sprain	Peptic ulceration
	Intercostal muscle strain	Gastroesophageal reflux
Referred pain from the thoracic spine	Rib stress fracture	Pneumothorax
Sternoclavicular joint disorders	Fractured sternum	Herpes zoster
Side strain		Pulmonary embolism

thoracic spine. Thus, patients presenting with anterior or lateral chest wall pain require a thorough examination of the cervicothoracic spine and thorax.

A list of possible causes of chest pain in athletes is presented in Table 28.2. The surface anatomy of this region is shown in Figure 28.13.

It may be difficult to distinguish between chest pain of cardiac origin and pain referred from the thoracic spine. They may both be unilateral and related to exercise. The clinical features of these two causes of chest pain in the athlete are considered in Table 28.3.

Major trauma to the chest wall is a medical emergency. Injuries sustained in contact sport commonly affect the ribs, resulting in either bruised or fractured ribs. These may lead to secondary dysfunction of the thoracic zygapophyseal joints, which can cause persistence of pain. Sternoclavicular and costochondral joint injuries

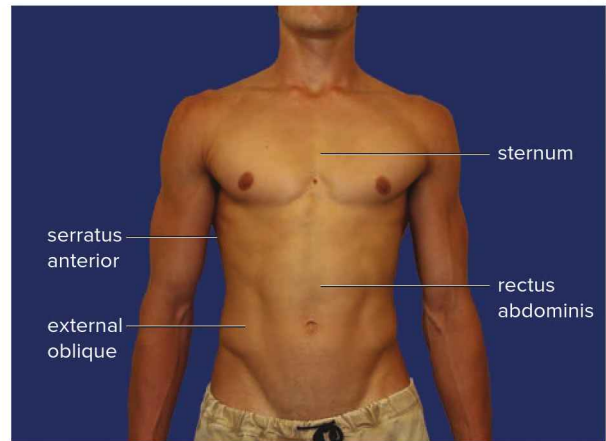


Figure 28.13 Surface anatomy of the anterior chest to assess for symmetry

Table 28.3 Comparison of clinical features of chest pain of cardiac origin and chest pain referred from the thoracic spine

Feature	Referred pain from thoracic spine	Myocardial ischaemia
Age	Any age, especially 20–40 years	Older (>35 years), greater likelihood with greater age
History of injury	Sometimes	No
Site and radiation	Spinal and paraspinal, arms, lateral chest, anterior chest, substernal, iliac crest	Retrosternal, parasternal, jaw, neck, inner arms, epigastrium, interscapular
Type of pain	Dull, aching, occasionally sharp, sudden onset and offset, severity related to activity, site and posture	Constricting, vice-like ('clenched-fist' sign), may be burning, gradual onset and offset
Aggravation	Deep inspiration, postural movement of thorax, certain activities (e.g. slumping or bending, walking upstairs, lifting, sleeping or sitting for long periods)	Exercise, activity, heavy meals, cold, stress, emotion
Relief	Maintaining erect spine, lying down, firm pressure on back (e.g. leaning against wall)	Rest Glyceryl trinitrate (GTN)
Associations	Chronic poor posture, employment requiring constant posture such as at a keyboard or computer	Cardiac risk factors such as family history, obesity, smoking, dyspnea, nausea, tiredness, pallor, sweating, vomiting

are not uncommon. Intercostal muscle strains have been considered a cause of chest wall pain but, on close clinical examination, many patients with this presentation are actually suffering referred pain from the thoracic spine. Stress fractures of the ribs occur in sports such as rowing, tennis, golf, gymnastics and baseball pitching where rapid torsion loads are common. The aetiology of chest wall muscle injury from violent overload is exemplified by cricket fast bowlers who induce a rapid alternation between muscle lengthening followed by rapid eccentric contraction during delivery. This action can result in a strain injury at the rib or costal cartilage insertion of the internal oblique muscle over the lower four ribs.

Clinical assessment

History

A history of trauma to the chest wall will lead the clinician to suspect a rib injury. In the absence of trauma, the history should distinguish between musculoskeletal conditions and other cardiac, gastrointestinal and respiratory causes. It is important to elicit the type of pain and the location of the pain. Associated symptoms such as palpitations, shortness of breath and sweating may indicate that the pain is cardiac in nature. A history of productive cough may suggest pain is of respiratory origin, while symptoms of food-related pain or a burning, boring pain behind the sternum should alert the clinician to include gastroesophageal reflux or peptic ulceration in the differential diagnosis.

Pain aggravated by deep inspiration or coughing may be musculoskeletal in nature or indicative of a respiratory problem. Associated thoracic and, to a lesser extent, cervical or lumbar pain may suggest the thoracic spine or rib joints as possible sources of the patient's chest pain. An increase in pain with trunk rotation might add to this suspicion.

Examination

Examination of the patient with chest wall pain should include palpation of the painful area and of possible sites of referral, especially the thoracic spine. Examination of the thoracic spine has been described earlier in this chapter. The cardiovascular and respiratory systems must always be examined. The abdomen should be examined for sources of referred pain.

Investigations

If there has been significant rib trauma, a chest radiograph should be performed to exclude the presence of a pneumothorax. Specific rib views may be necessary to detect rib fractures. Chest radiography will indicate cardiac size and may reveal evidence of respiratory infection.

MRI (nuclear imaging/CT where MRI not available) may be required to detect stress fractures or other reactive bony conditions.

Electrocardiography and other cardiac investigations, including a stress ECG and an echocardiograph, may be performed if there is clinical suspicion of cardiac dysfunction. A Holter monitor may be indicated to help detect occult cardiac pacing irregularities. Gastroscopy may be indicated if peptic ulceration or reflux is suspected.

Rib trauma

A direct blow to the chest may result in trauma to the ribs. This may range from bruising to an undisplaced or displaced rib fracture or to costal cartilage displacement (Fig. 28.14). Typically, the patient complains of pain aggravated by deep inspiration or coughing. Examination reveals local tenderness over one or more ribs.

A pneumothorax or rarely, a haemopneumothorax, may occur as a result of a rib fracture. Any athlete with rib trauma must undergo a respiratory examination to exclude these conditions. It is also important to consider trauma to underlying structures such as the liver, spleen and kidneys.

Radiographs may confirm the presence of a rib fracture, although it is not essential as treatment is symptomatic. Injury to the upper four ribs is unusual as they are somewhat protected. The lower two ribs are likewise rarely fractured as they are not attached to the sternum.

Treatment consists of analgesia and encouragement of deep breathing to prevent localised lung collapse. A fractured rib can be extremely painful. It will continue to be painful and tender to palpation for at least 3 weeks. Bruised ribs may also be painful and tender for up to 3 weeks.

Return to sport for athletes with an undisplaced rib fracture is appropriate when pain settles. Protective padding or splints may be used in contact sports after a rib

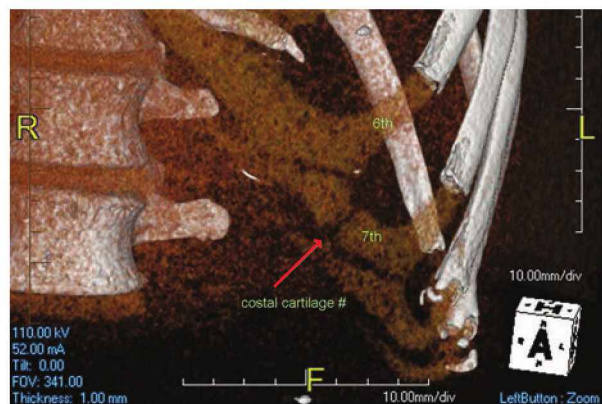


Figure 28.14 Fracture of the costal cartilage in a 30-year-old footballer

injury. Local anaesthesia may be used for short-term pain management where indicated and with due care to avoid causing a pneumothorax. Contusion of the costochondral joints can be painful. Local treatment consists of cold therapy and strapping to splint the region.

Referred pain from the thoracic spine

Referred pain to the chest may or may not be associated with a history of thoracic spine pain. On examination, there will usually be marked tenderness and stiffness, either centrally over the spinous processes of the thoracic vertebrae or, more commonly, on the same side as the chest pain over the thoracic zygapophyseal or costovertebral joints. There is also often associated tenderness in the soft tissues surrounding the thoracic spine, especially the paravertebral muscles. Active trigger points may develop and contribute to the referred pain. Referred pain, however, may not follow predictable patterns, which requires the clinician to explore symptoms and to rule out visceral disorders.^{5, 13}

Local treatment aims to restore full range of motion of the involved thoracic intervertebral segments by mobilisation or manipulation. Soft tissue therapy to surrounding areas may be helpful. This includes digital ischaemic pressure to painful sites, and sustained myofascial tension and other massage techniques where chronic muscle and soft tissue tightness is established. Dry needling of involved trigger points may also be considered.

PRACTICE PEARL

Referred pain from the thoracic spine is probably the most common cause of chest wall pain in the young athlete.

Sternoclavicular joint problems

The sternoclavicular (SC) joint is the sole articulation between the upper extremity and the axial skeleton. The joint itself is diarthrodial, with an articular disc interposed between the two bones (Fig. 28.15). The articular surface of the medial clavicle is much greater than the sternal articulation. Only about 25% of the medial clavicle's surface articulates with the sternum at any one time, making the SC joint the joint with the least bony stability in the body.

The integrity of the joint comes from the strong surrounding ligaments, including the anterior, posterior, superior and inferior SC ligaments and the interclavicular, costoclavicular and intra-articular disc ligaments. The epiphysis of the medial clavicle is the last to ossify and fuse, at around 18 and 25 years of age, respectively. Another feature of this joint is the number of vital structures located directly posterior to the joint. These include the subclavian veins and artery, the trachea, the oesophagus and the mediastinum.

The SC joint can be injured by a direct blow or, more commonly, indirectly from a blow to the shoulder and is

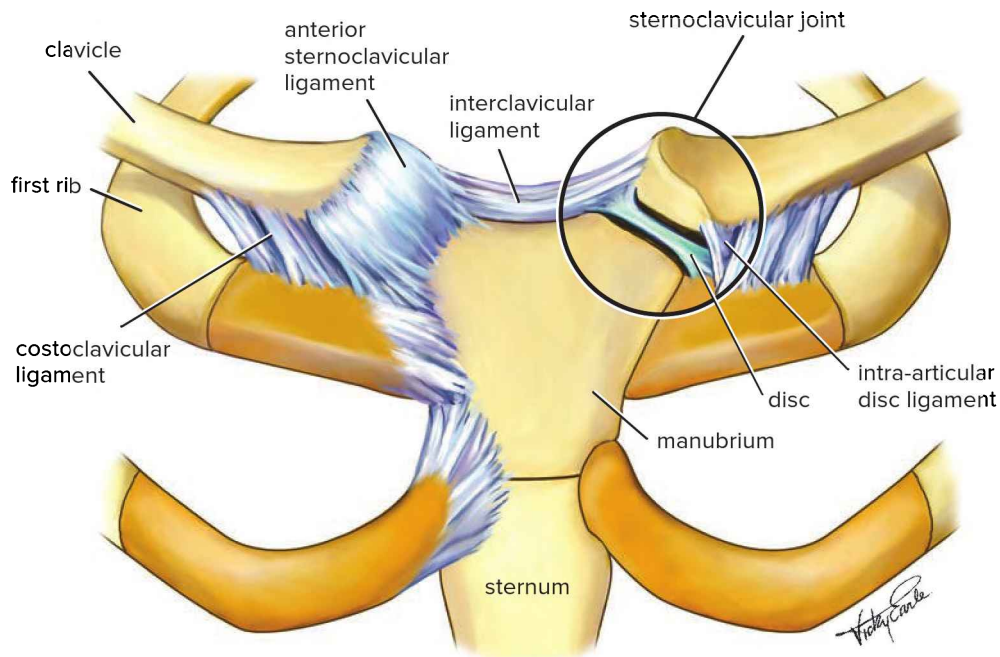


Figure 28.15 Anatomy of the sternoclavicular joint

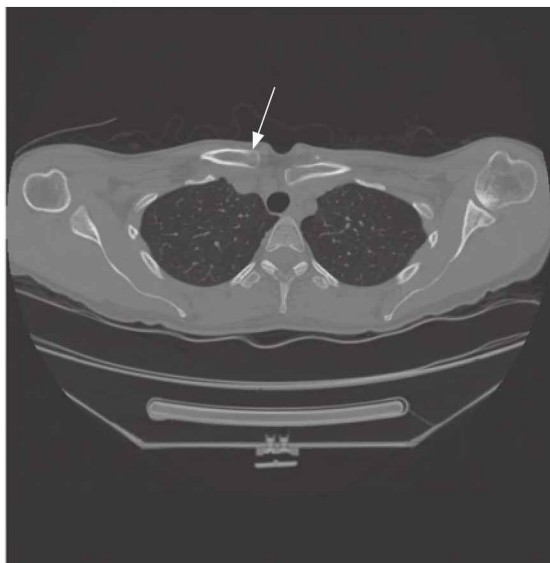


Figure 28.16 CT scan of anterior sternoclavicular joint dislocation showing clear asymmetry between the dislocated side (arrow) and the normal articulation

seen in upper limb weight-bearing sports (e.g. gymnastics). Simultaneous injuries of the acromioclavicular (AC) and SC joints are reasonably common.^{18–20}

Traumatic injuries of the SC joint can be divided into first and second-degree sprains involving the joint capsule, subluxations, and dislocations involving rupture of the SC and/or costoclavicular ligaments and fractures of the medial clavicle. Subluxations and dislocations are further divided into anterior (more common; Fig. 28.16) and posterior (more dangerous given the adjacent neurovascular structures).^{21–23}

Patients present with local pain and swelling depending on the severity of the injury. Anterior subluxations can be treated symptomatically by a figure-of-eight bandage for 1–2 weeks. Anterior dislocations can be reduced with lateral traction on the abducted arm and direct pressure over the medial clavicle. Dislocations are immobilised for 3–4 weeks in a clavicular strap. If the joint redislocates after closed reduction, a period of observation is appropriate. Many patients remain asymptomatic.

Posterior dislocations should be reduced closed under general anaesthesia as soon as possible after the injury. Traction is applied to the patient's abducted arm and the medial clavicle is brought forward manually or with a towel clip. Reductions are generally stable and are held with a figure-of-eight strap for 4 weeks.

Costochondritis

Costochondritis occurs at the plane joints between the sternum and ribs. It is characterised by activity-related

pain and tenderness localised to the costochondral junction, usually over the T2–5 segments. Tietze's syndrome describes a painful inflammation of usually a single costochondral joint, although the sternoclavicular joint may be involved.²⁴

Treatment consists of nonsteroidal anti-inflammatory drugs (NSAIDs), local physiotherapy and gentle mobilisation of the costochondral joints. This can prove an extremely difficult condition to treat. Corticosteroid injection to the costochondral junction may be of some assistance in refractory cases.

Stress fracture of the ribs

Stress fracture of the ribs has been reported with a number of sports and is due to excessive muscle traction at the attachments to the ribs.^{25–28} Stress fracture of the first rib is seen in baseball pitchers and appears to occur at the site of maximal distraction between the upward and downward muscular forces on the rib. This stress fracture tends to heal poorly. The injury has also been reported in golfers, generally on the leading side (i.e. left ribs in a right-handed player).

Anterolateral stress fractures of the ribs (Fig. 28.17), mainly the fourth and fifth ribs, are commonly seen in rowers. This stress fracture is thought to be due to excessive action of the serratus anterior muscle. The biomechanics of the rowing action (Chapter 9) should be assessed and discussed with the coach and athlete to determine the possible cause/s and changes to technique where indicated. The concept of excess rib cage

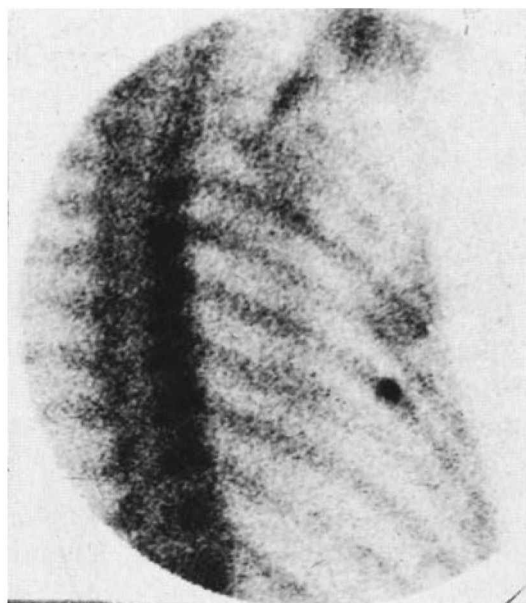


Figure 28.17 Isotopic bone scan of a stress fracture of the ribs

compression coupled with excessive isometric thoracic muscle contraction in the beginning of the drive phase in rowing has been proposed as the mechanism.²⁸⁻³⁰

Side strain

with ANDREW NEALON

'Side strain' is an injury sustained by athletes that is characterised by pain and focal tenderness of the lateral trunk over the inferior rib cage.^{31, 32} It has most commonly been reported in cricket fast bowlers^{31, 33-35} and baseball players,^{32, 36, 37} and has also been reported in case studies in a golfer, javelin thrower, rower,³¹ rugby player³⁵ and professional tennis player.³⁸ The injury typically occurs on the side contralateral to the bowling arm in fast bowlers.³¹ In major league baseball the majority of injuries occur to the side contralateral to the throwing arm in pitchers, and the side facing the pitcher in batters.³² It is evident that side strain is most commonly reported in sports that require highly repetitive, unilateral, asymmetrical and explosive actions that place great demand on the trunk.³⁹

Clinical features

The athlete typically presents with sudden pain and focal tenderness over the lateral aspect of the lower costal margins,^{31-32, 35-38} and may initially have considerable pain to breathe deeply,^{31, 36, 38} cough and roll over in bed.³⁹ Palpation is typically reported to reproduce pain. Diagnosis is primarily clinical, although imaging with MRI or ultrasound is common practice in elite sport.³⁹

Pain is commonly reproduced by movements associated with the initial mechanism of injury.³¹ Trunk lateral flexion range of motion is a common clinical test and is typically reduced and limited by pain when moving both towards and away from the injury side.³⁹ Isometric resisted shoulder adduction on the injury side may also reproduce symptoms.³⁹

Pathology and epidemiology

The pathology of side strain has been described in MRI³¹ and ultrasound³⁵ studies as a tear of the internal oblique muscle from its rib or costal cartilage attachment with acute oedema and haemorrhage (Fig 28.18). Rib oedema and stripping of the rib periosteum have also been reported in a significant proportion of cases.³¹ This suggests that some side strain injuries involve both soft tissue and bony pathology though the effect of this on recovery time is yet to be determined.³⁹ Injury from the undersurface of the eleventh rib is most common, followed by the tenth and then ninth ribs.³¹ The relationship between imaging findings and recovery time is yet to be determined and is the subject of current research.

PRACTICE PEARL

Cricket fast bowlers are particularly vulnerable to side strain injury, with over half of the fast bowlers playing regularly at first class level in Australia and England reporting at least one episode during their career.

The injury tends to occur most commonly in the early twenties and often during the early stages of a player's career at first class level. Side and abdominal strain contributed the second highest incidence of all body areas over the past two decades in the most recent Australian first class cricket injury report, causing the third highest total player unavailability behind only lumbar stress fractures and thigh and hamstring strains.⁴⁰ It also comprised 5% of all injuries in major league baseball.³² Injury surveillance studies in both cricket and baseball suggest that the incidence of side strain injury has increased in recent years.^{32, 40}

The mechanism of side strain injury is difficult to establish, although it is typically incurred in a single specific activity, such as bowling or throwing, and prevents further play.³⁹ It has been postulated to occur due to sudden eccentric contraction of the internal oblique muscle after the lengthening that occurs during the wind-up phase of bowling or throwing.³¹ However, trunk lateral flexion range of motion during the fast bowling action exceeds the range demonstrated in standing by a mean of 29%⁴¹

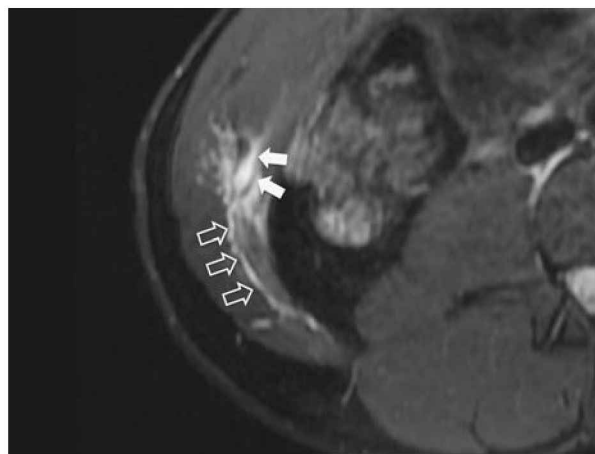


Figure 28.18 MRI (axial proton density image with fat saturation) of a side strain injury of the right internal oblique muscle. Focal grade II tearing of the internal oblique muscle is shown (solid arrows) just below the muscle attachment to the periosteum of the undersurface of the tenth rib. Intermuscular haemorrhage (open arrows) tracks between the internal and external oblique muscles

and compression of the lower ribcage versus the iliac crest may also be a factor. Side strain is most commonly sustained during the early stages of the competitive playing season.^{32, 42} This suggests that adjusting to the demands of competition intensity may be a risk factor for this injury.³⁹

Management

Treatment includes resting until pain has resolved during respiration followed by the restoration of strength and lateral flexion mobility of the trunk with graduated return to sports-specific activity. The repetitive nature of the sports commonly associated with side strain mean that high volumes of strengthening exercise need to be performed in the rehabilitation phase so that the athlete can cope with the demands of sport upon returning to play.³⁹

Recovery time can be prolonged. Case series of cricket and baseball players report times of 4–10 weeks to return to play.³⁹ Currently ongoing research in first class cricket in Australia and England indicates that while about 20% of first class cricket fast bowlers return to pre-injury level of competition within 3 weeks, almost 50% require more than 5 weeks. In excess of 7 weeks was required for over 20% of fast bowlers. Major league baseball surveillance reports that baseball players spent a mean time of 4–5 weeks on the disability list over the 20 years to 2010.³²

PRACTICE PEARL

Recurrence is common—30% of cricket first class bowlers suffered a further injury.

This is most common in the first 12 months, with 20% of all cases recurring in this time. In the only published

case series of first class cricket fast bowlers to date, six out of ten cases were recurrences.³³ Baseball data reports a lower recurrence rate of 12%.³²

Costoiliac impingement symptoms following side strain injury are also common, experienced by over a third of fast bowlers upon their return to bowling during rehabilitation and persistent in 20% of cases at time of return to competition. Currently it is unknown if such symptoms are associated with risk of recurrence. Non-remitting pain has led to surgery in cricket fast bowlers, with excision of the involved rib resolving symptoms.³⁴

Despite its high prevalence in sports such as cricket and baseball, knowledge on side strain remains limited. The relationship between clinical assessment measures, MRI findings and time to return to play in first class cricket fast bowlers is the subject of current research.

Conclusion

Disorders of the thoracic region in athletes can be complex given the interactions between musculoskeletal, cardiorespiratory and visceral systems, and their convergence in innervation patterns. Athletic training and competition impose unique strains on the thorax and thoracic spine, particularly during the important skeletal development years, which may give rise to a variety of stress or overuse responses.

In considering a differential diagnosis, the clinician needs to consider the biomechanics of the activity and common soft tissue and joint disorders of the thoracic region, and to be alert to the potential for occult symptoms masquerading as musculoskeletal problems. Similarly, symptoms suggestive of a musculoskeletal presentation may indeed arise from cardiac or visceral systems which can require timely referral and intervention.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e